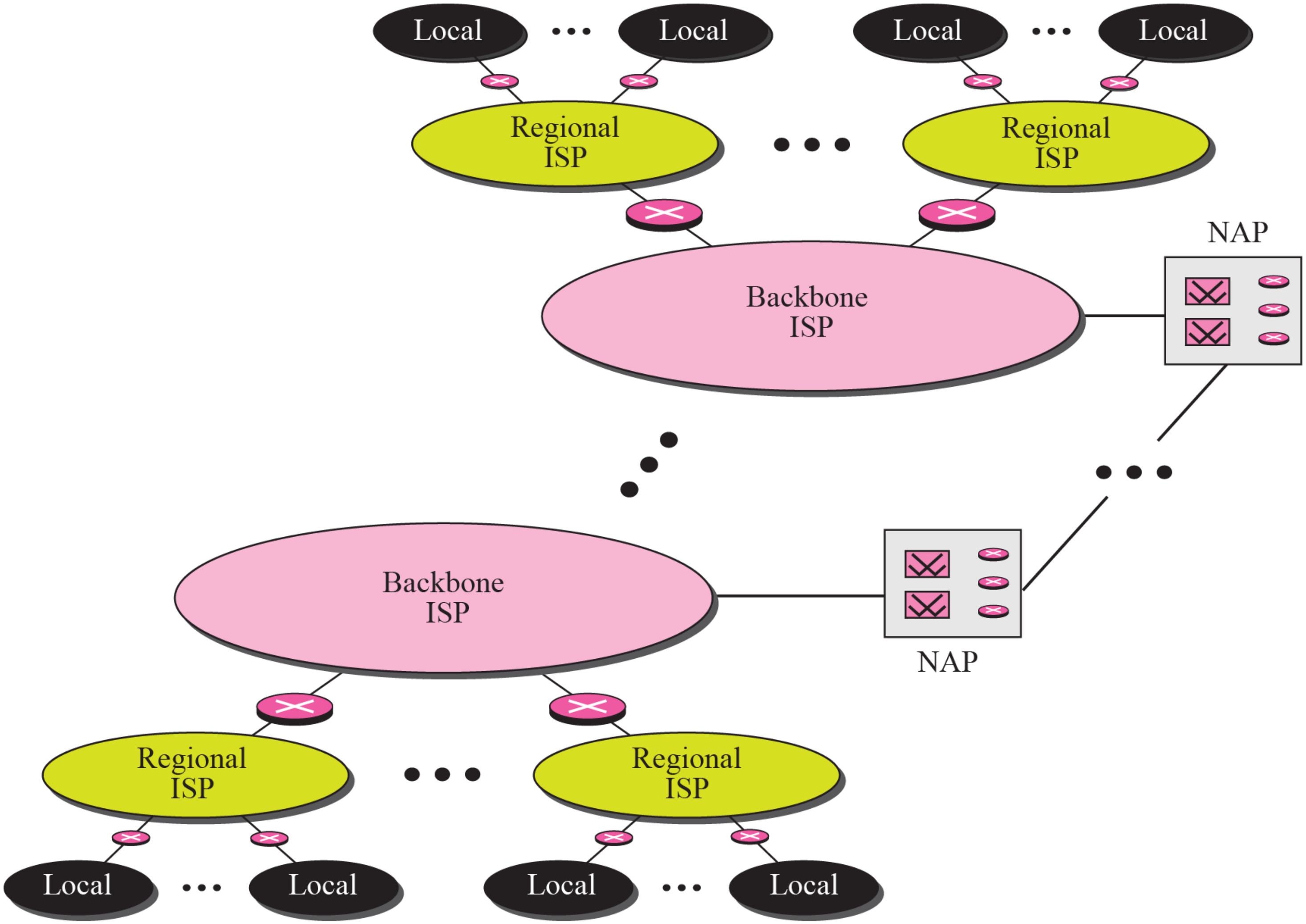
EETS 7302 TCP/IP Network Administration

Session 1
Introduction

Internet today

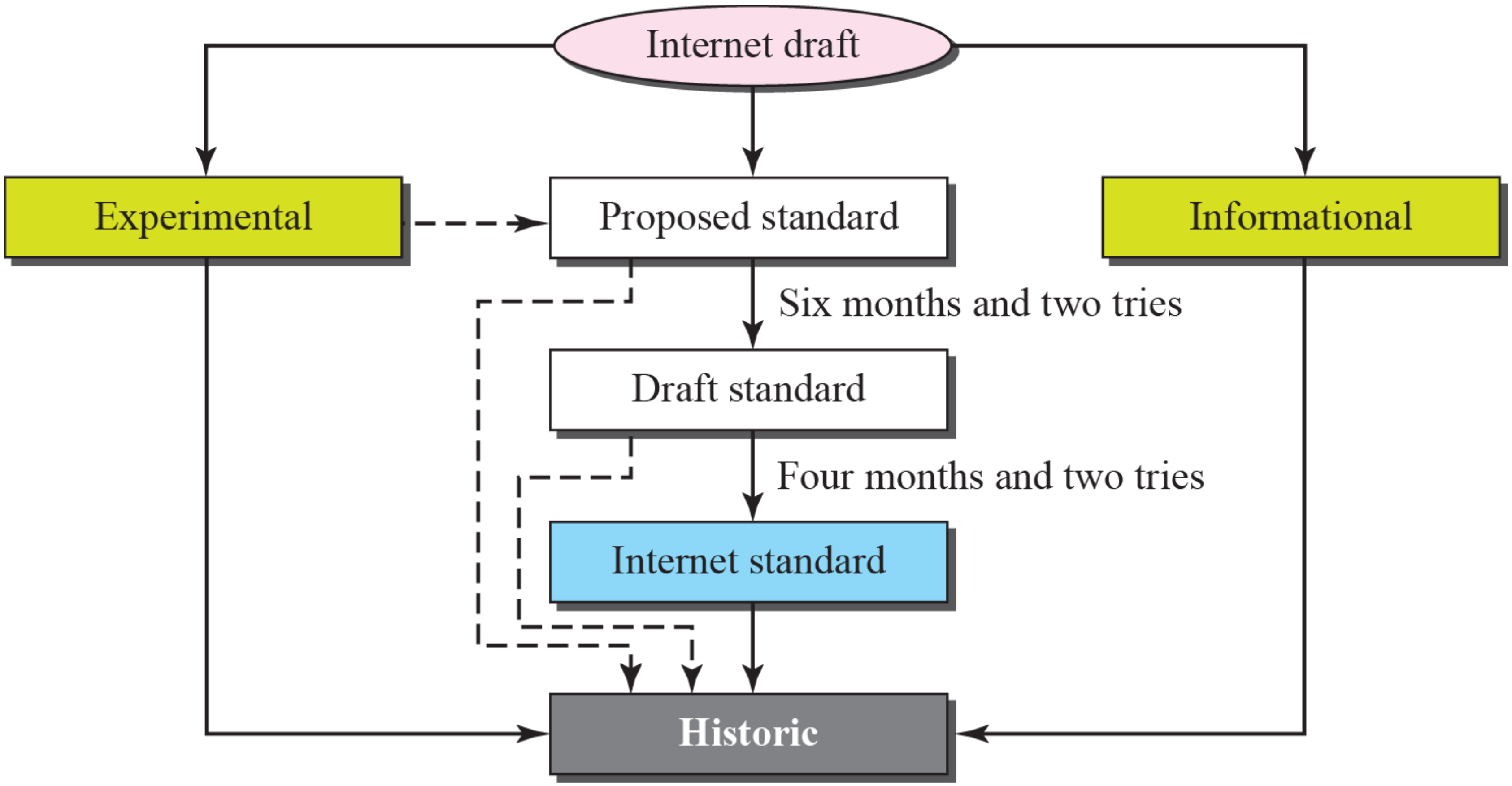


STANDARDS ORGANIZATION

Standards **are developed through the cooperation of standards creation committees, forums, and government regulatory agencies.**

INTERNET STANDARDS

- An Internet standard is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet.
- It is a formalized regulation that must be followed.
- There is a strict procedure by which a specification attains Internet standard status.
- A specification begins as an Internet draft.
- An Internet draft is a working document with no official status and a six-month lifetime.



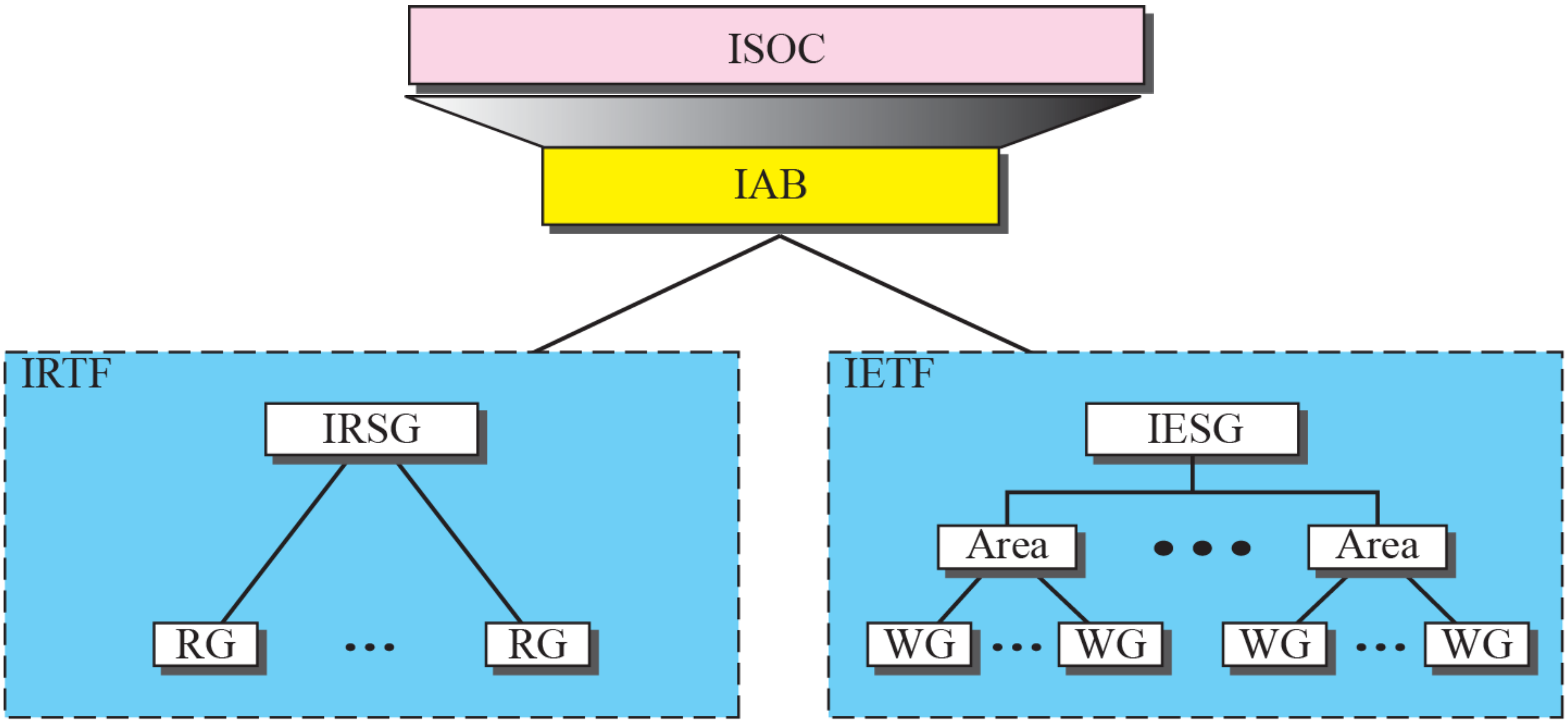
RFCs can be found at
http://www.rfc-editor.org.

INTERNET ADMINISTRATION

- The Internet, with its roots primarily in the research domain, has evolved and gained a broader user base with significant commercial activity.
- Various groups that coordinate Internet issues have guided this growth and development.

Internet administration

- ✓Internet Society (ISOC)
- ✓Internet Architecture Board (IAB)
- ✓Internet Research Task Force (IRTF)



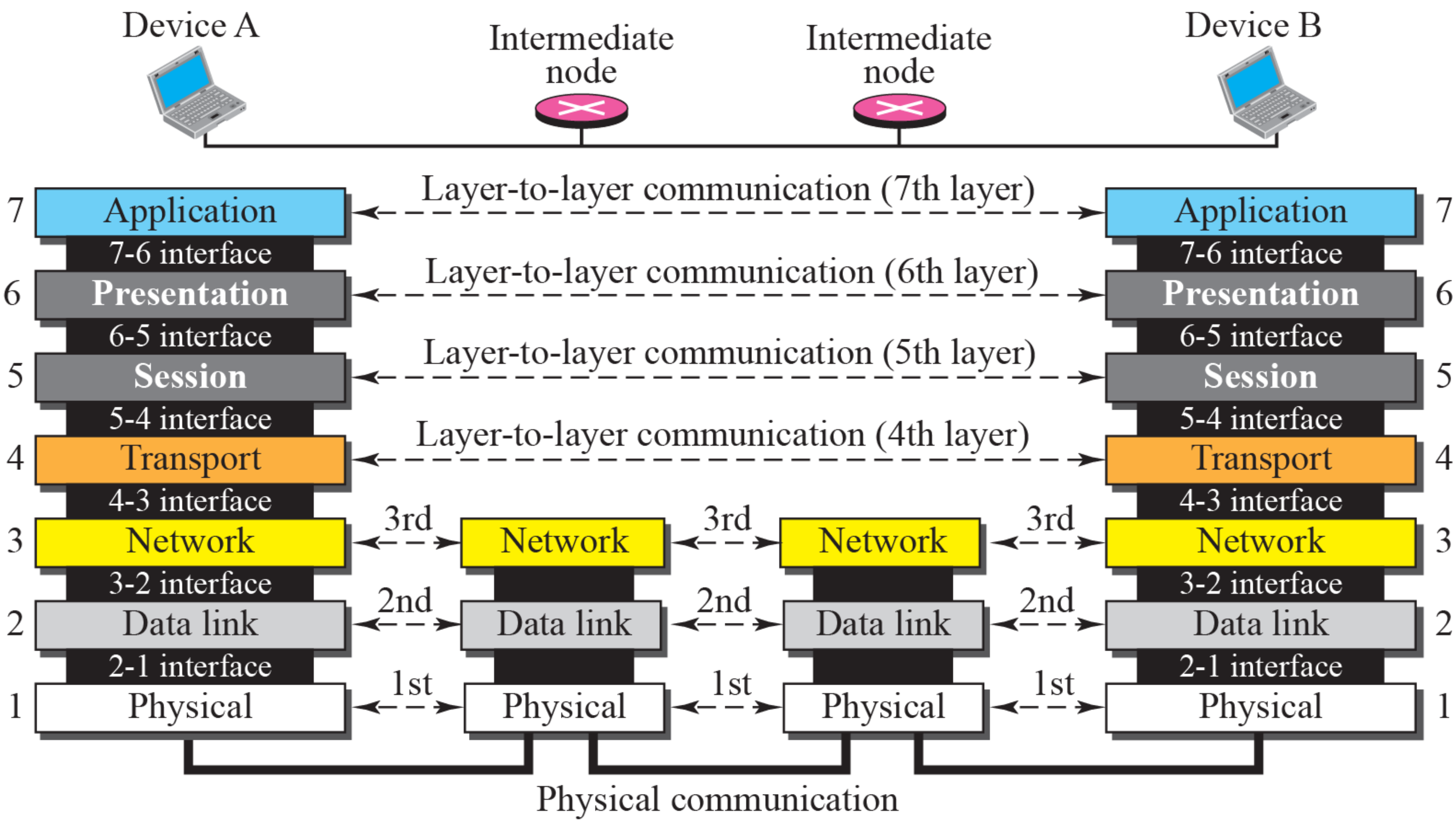
PROTOCOL LAYERS

- **a protocol is required when two entities need to communicate.**
- **When communication is not simple, we may divide the complex task of communication into several layers.**
- **we may need several protocols, one for each layer.**

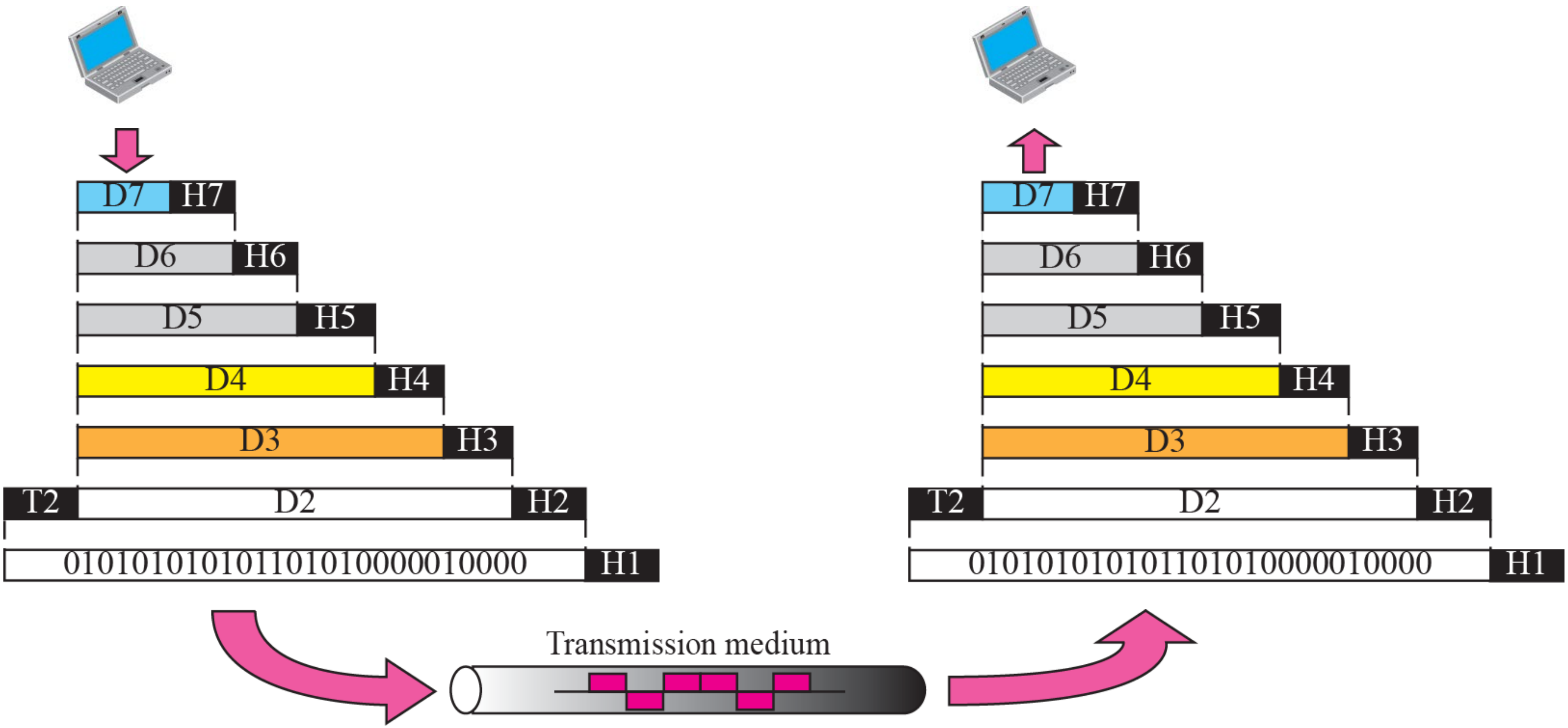
THE OSI MODEL

- **Established in 1947, the *International Standards Organization (ISO)* is a multinational body dedicated to worldwide agreement on international standards.**
- **Almost three-fourths of countries in the world are represented in the ISO.**
- **An ISO standard that covers all aspects of network communications is the *Open Systems Interconnection (OSI)* model.**
- **It was first introduced in the late 1970s.**

OSI layers


Device A
Intermediate node
Intermediate node
Device B
7 Application
7-6 interface
6 Presentation
6-5 interface
5 Session
5-4 interface
4 Transport
4-3 interface
3 Network
3-2 interface
2 Data link
2-1 interface
1 Physical
Layer-to-layer communication (7th layer)
Layer-to-layer communication (6th layer)
Layer-to-layer communication (5th layer)
Layer-to-layer communication (4th layer)
3rd
2nd
1st
Network
Data link
Physical
Network
Data link
Physical
Application 7
7-6 interface
Presentation 6
6-5 interface
Session 5
5-4 interface
Transport 4
4-3 interface
Network 3
3-2 interface
Data link 2
2-1 interface
Physical 1
Physical communication

An exchange using the OSI model



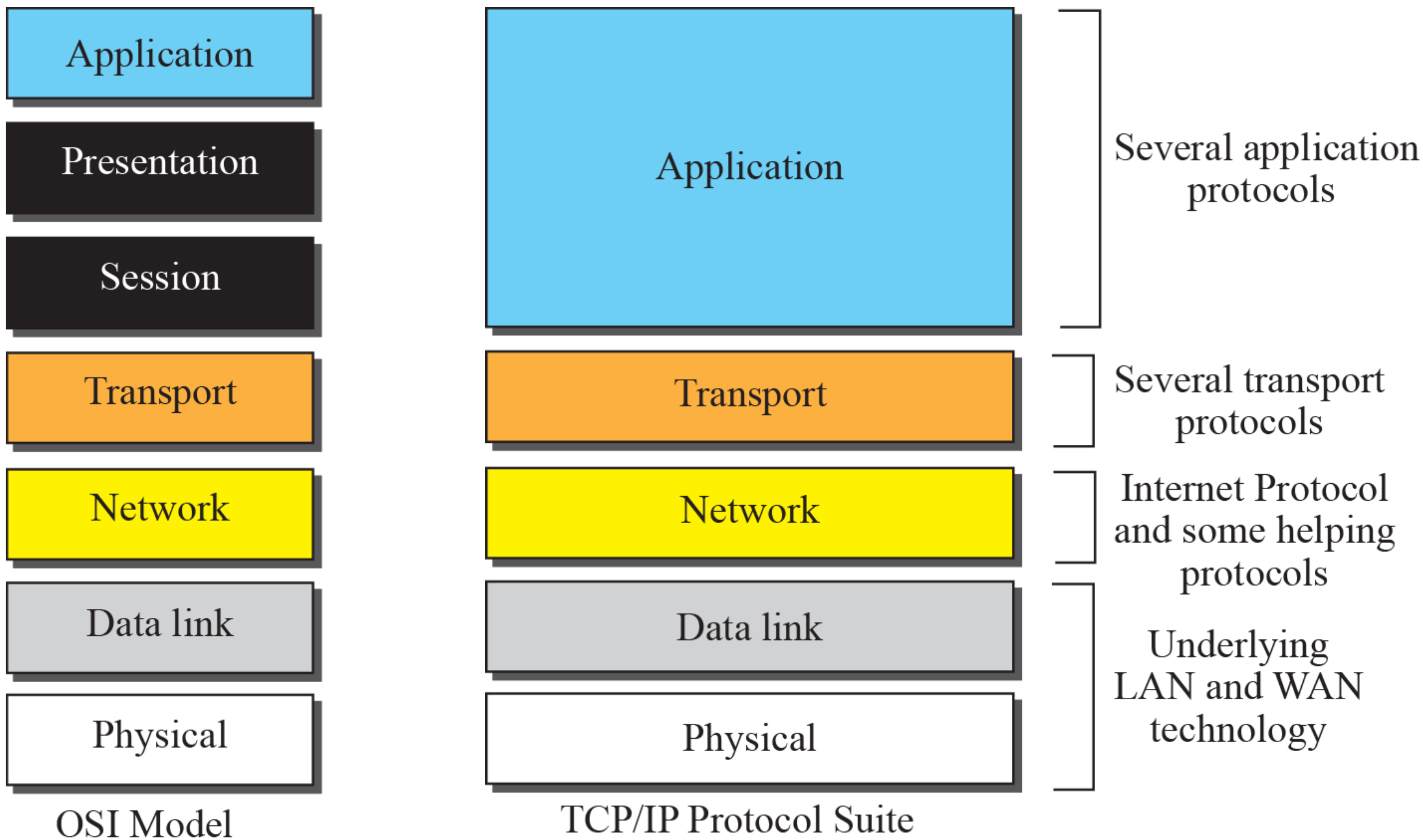
Summary of OSI Layers

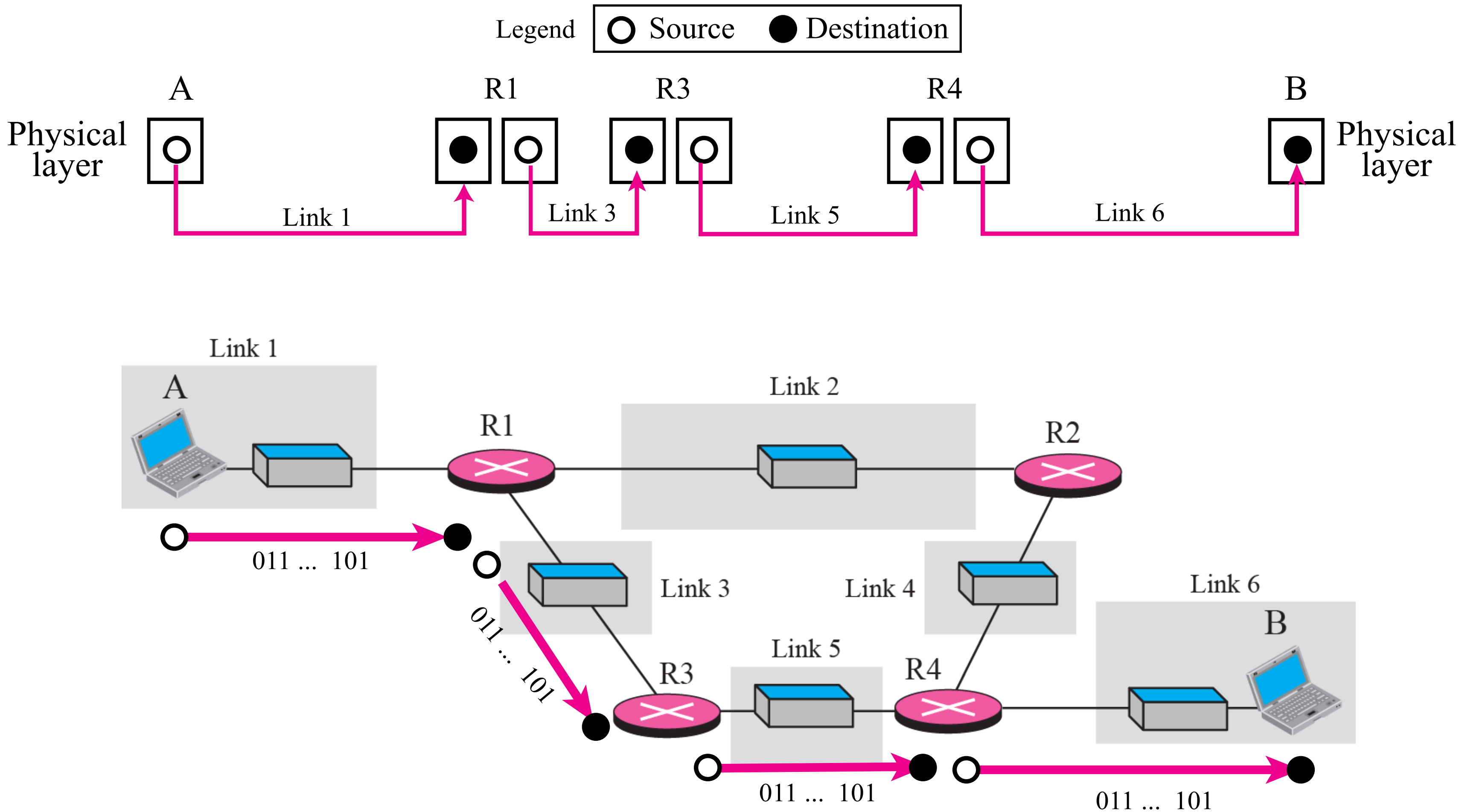
Application	To allow access to network resources	7
Presentation	To translate, encrypt, and compress data	6
Session	To establish, manage, and terminate sessions	5
Transport	To provide reliable process-to-process message delivery and error recovery	4
Network	To move packets from source to destination; to provide internetworking	3
Data link	To organize bits into frames; to provide hop-to-hop delivery	2
Physical	To transmit bits over a medium; to provide mechanical and electrical specifications	1

TCP/IP PROTOCOL SUITE

- The TCP/IP protocol suite was developed prior to the OSI model.
- Therefore, the layers in the TCP/IP protocol suite do not match exactly with those in the OSI model.
- The original TCP/IP protocol suite was defined as four software layers built upon the hardware.
- Today, however, TCP/IP is thought of as a five-layer model with the layers named similarly to the ones in the OSI model.

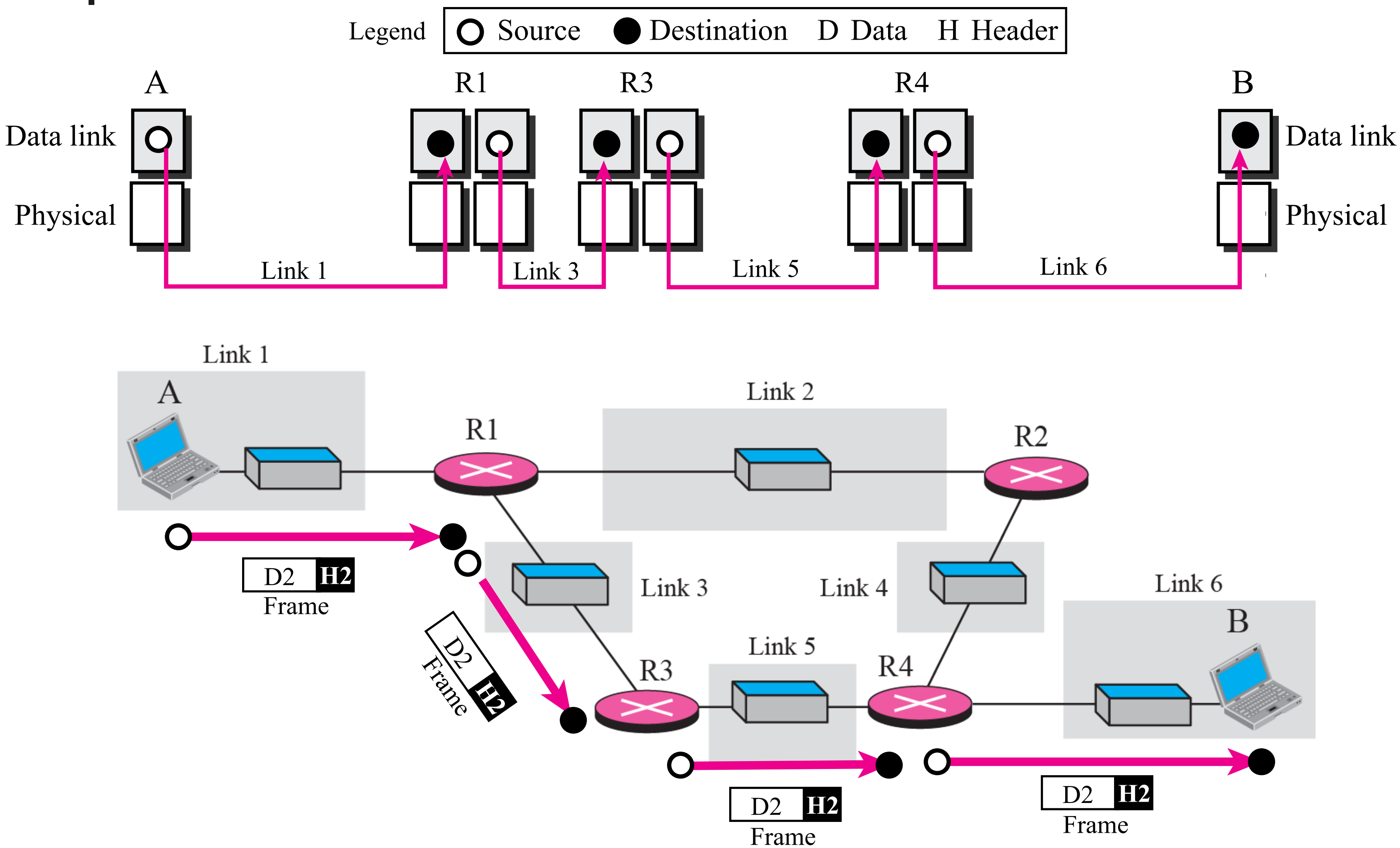
TCP/IP and OSI model





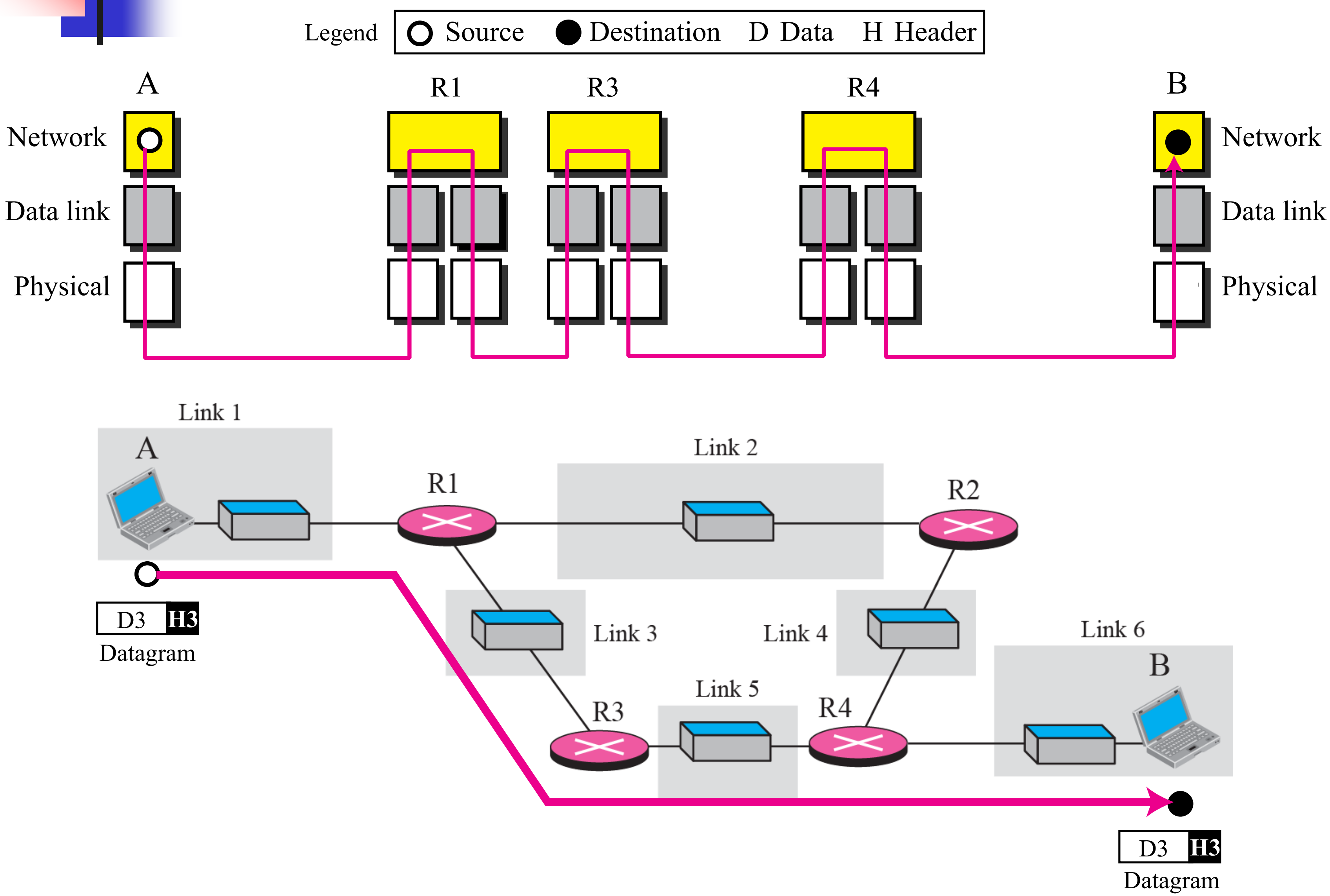
The unit of communication at the physical layer is a bit.

Communication at the data link layer



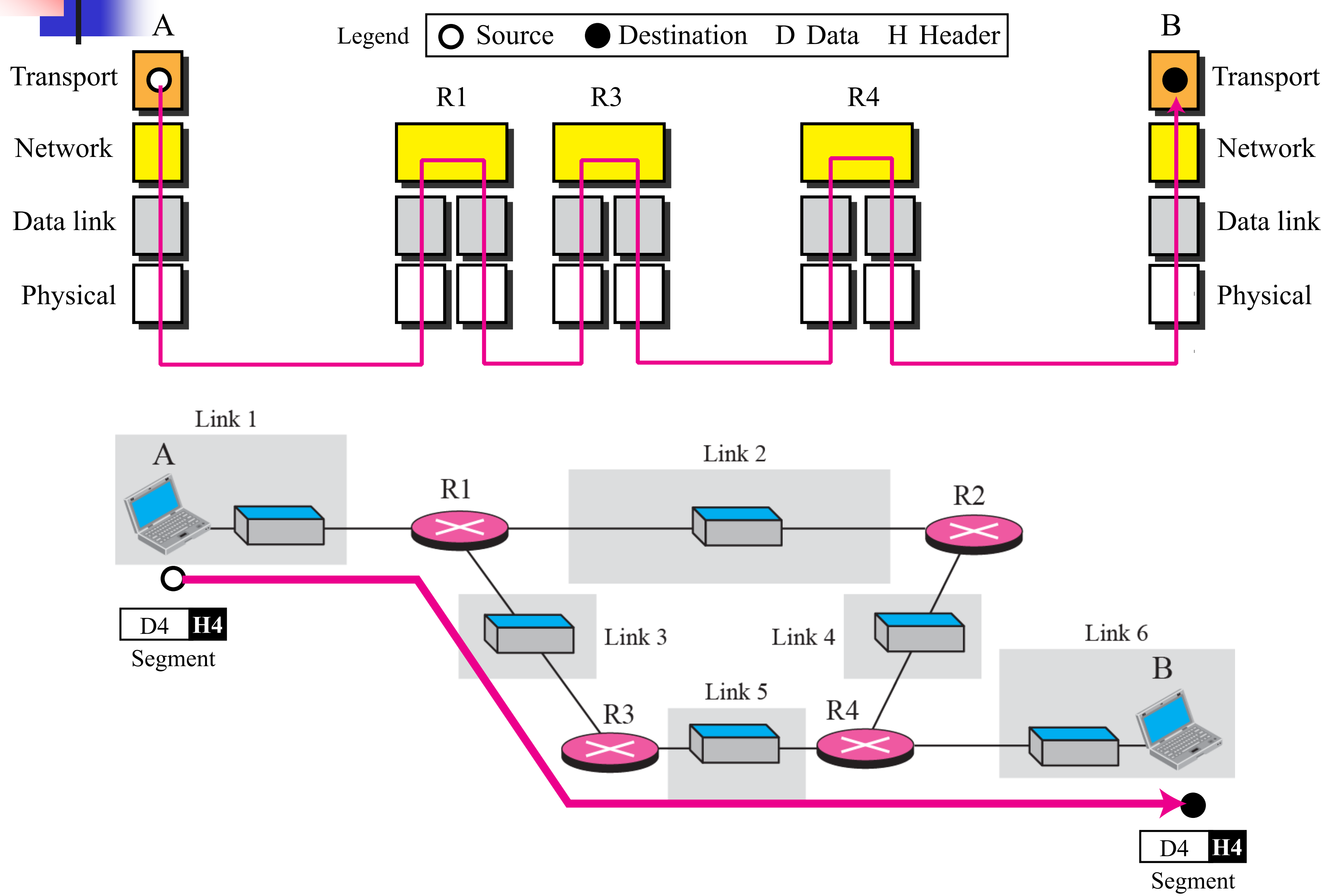
The unit of communication at the data link layer is a frame.

Communication at the network layer



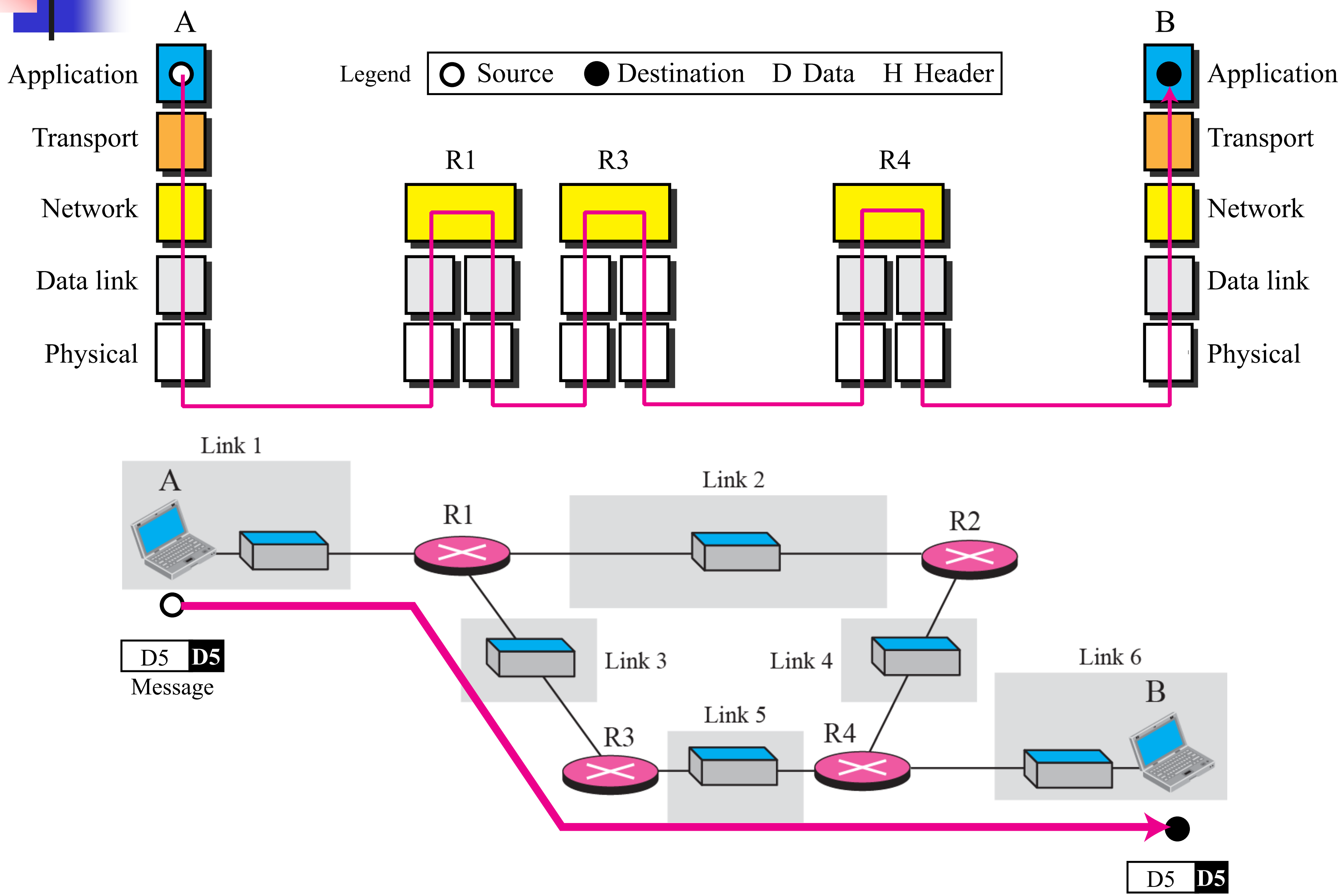
The unit of communication at the network layer is a datagram.

Communication at transport layer



The unit of communication at the transport layer is a segment, user datagram, or a packet, depending on the specific protocol used in this layer.

Communication at application layer

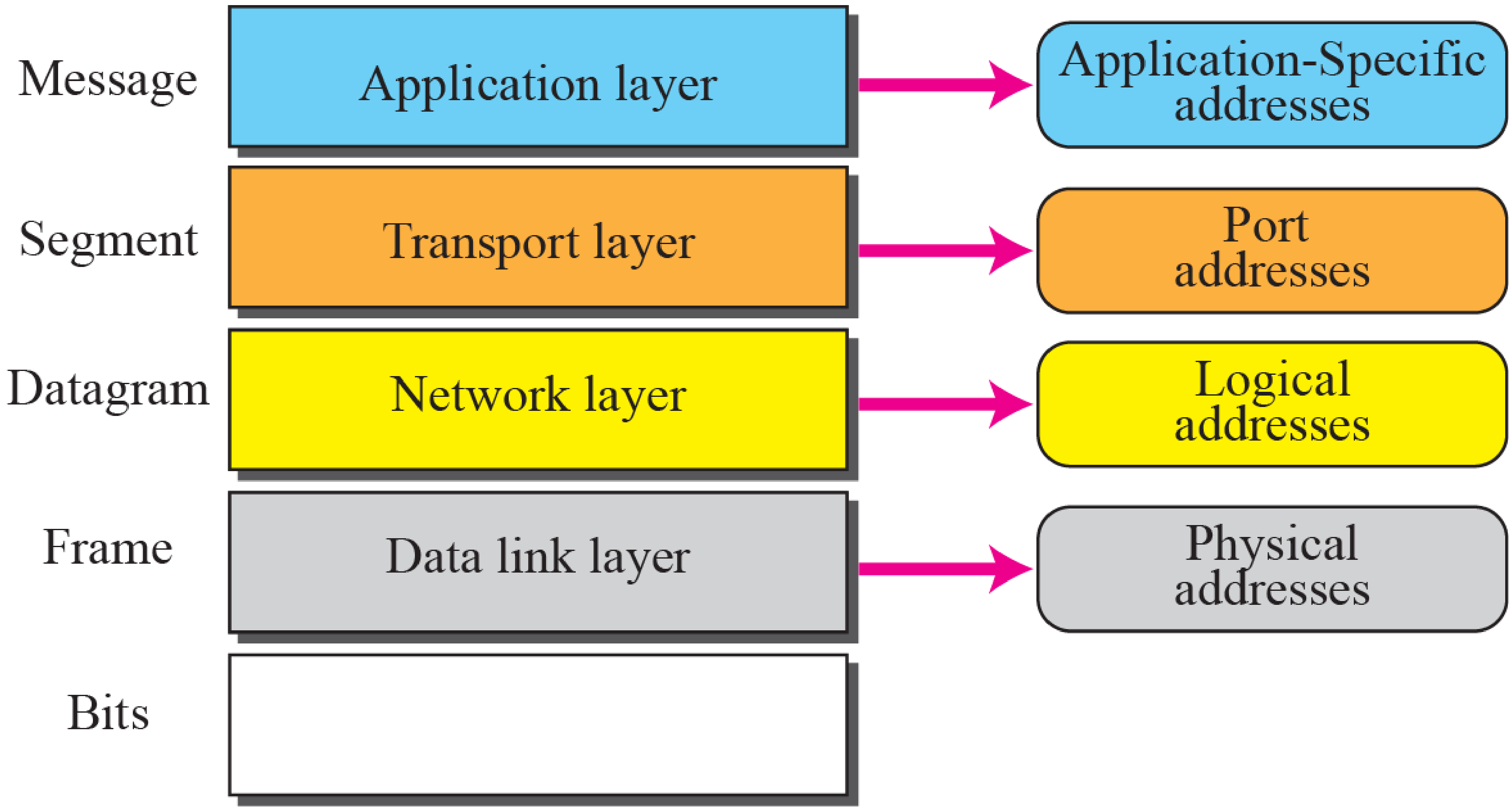


The unit of communication at the application layer is a message.

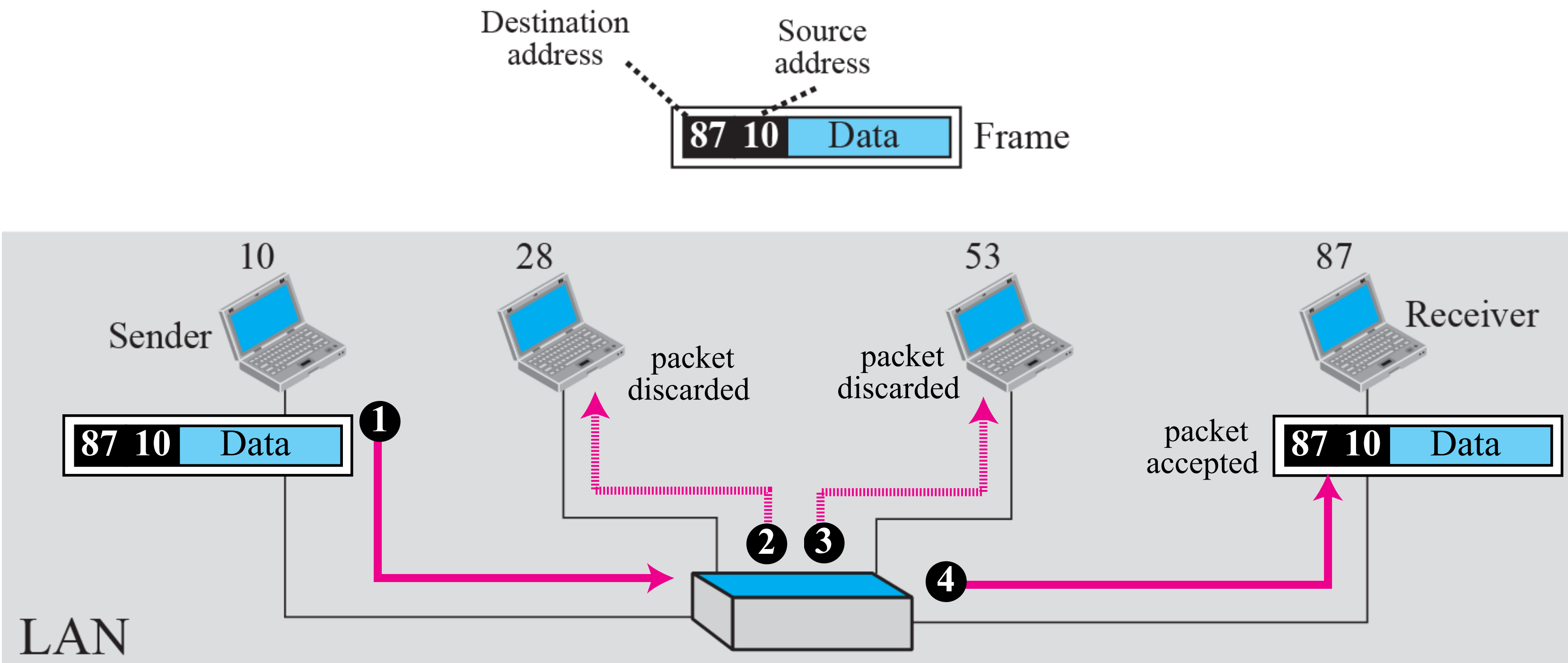
ADDRESSING

- Four levels of addresses are used in an internet employing the TCP/IP protocols:
  - physical address,
  - logical address,
  - port address,
  - application-specific address.
- Each address is related to a one layer in the TCP/IP architecture

Addresses in the TCP/IP protocol suite



Example 2.3: physical addresses



Example

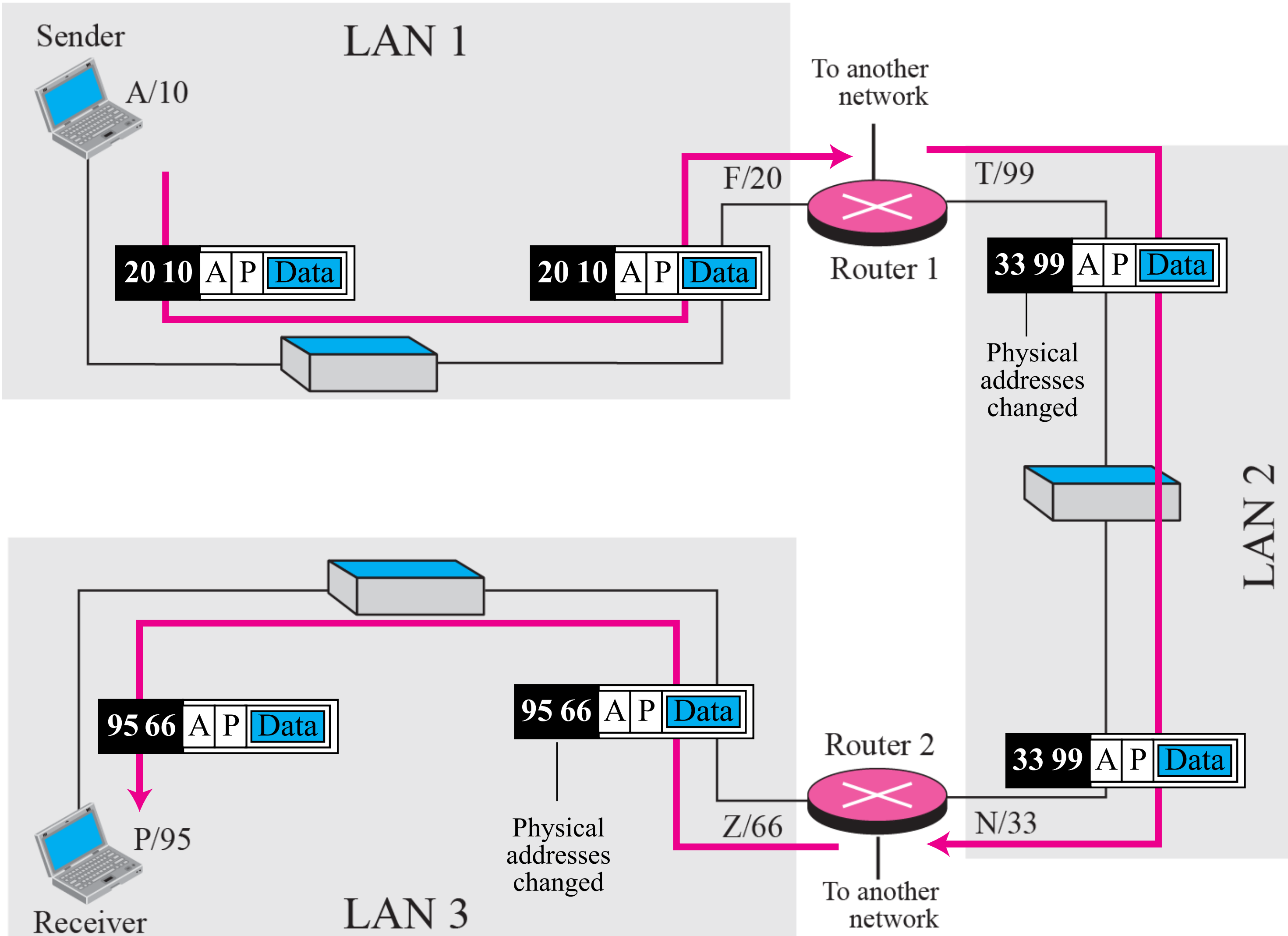
most local area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:

07:01:02:01:2C:4B

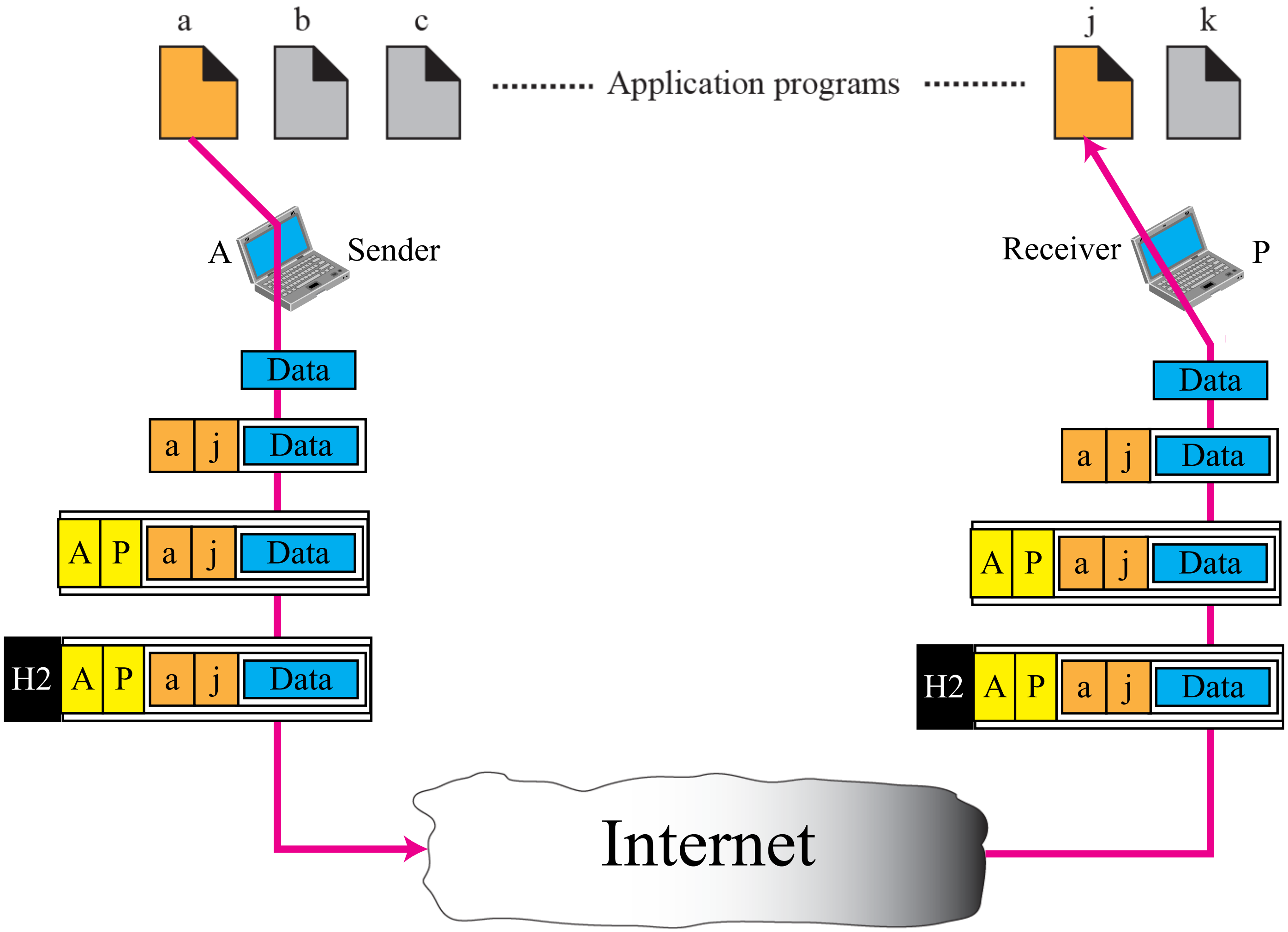
A 6-byte (12 hexadecimal digits) physical address

Example : logical addresses

The physical addresses will change from hop to hop, but the logical addresses remain the same.



The physical addresses change from hop to hop, but the logical and port addresses usually remain the same.



Example

a port address is a 16-bit address represented by one decimal number as shown.

753

A 16-bit port address represented as one single number

References

TCP/IP Protocol Suite (4th Edition),

by Behrouz A. Forouzan – Used the text's slide deck

End of session